

Cocoa, Blood Pressure, and Cardiovascular Health.



High blood pressure is an important risk factor for cardiovascular disease and cardiovascular events worldwide. Clinical and epidemiological studies suggest that cocoa-rich products reduce the risk of cardiovascular disease. According to this, cocoa has a high content in polyphenols, especially flavanols. Flavanols have been described to exert favorable effects on endothelium-derived vasodilation via the stimulation of nitric oxide-synthase, the increased availability of l-arginine, and the decreased degradation of NO. Cocoa may also have a beneficial effect by protecting against oxidative stress alterations and via decreased platelet aggregation, decreased lipid oxidation, and insulin resistance. These effects are associated with a decrease of blood pressure and a favorable trend toward a reduction in cardiovascular events and strokes. Previous meta-analyses have shown that cocoa-rich foods may reduce blood pressure. Long-term trials investigating the effect of cocoa products are needed to determine whether or not blood pressure is reduced on a chronic basis by daily ingestion of cocoa. Furthermore, long-term trials investigating the effect of cocoa on clinical outcomes are also needed to assess whether cocoa has an effect on cardiovascular events. A 3 mmHg systolic blood pressure reduction has been estimated to decrease the risk of cardiovascular and all-cause mortality. This paper summarizes new findings concerning cocoa effects on blood pressure and cardiovascular health, focusing on putative mechanisms of action and "nutraceutical " viewpoints.